Data Access and Evaluation Procedure

How to access the datasets

Access to ManGO

The dataset is stored on ManGO. Users can connect to ManGO by using different clients from any computer after KU Leuven IDP login.

The landing page for all ManGO clients is the ManGO portal. After logging in, you will get an overview of all zones you have access to. By clicking on 'Enter portal', you will go to the ManGO portal for that zone. If you prefer to access ManGO via a different client, you can find the necessary credentials or code under 'How to Connect'.

You can find more information in the documentation: https://docs.vscentrum.be/data/tier1data/clients/mango_portal.html.

Access to the data can also be done through CLI using iCommands (documentation: https://docs.vscentrum.be/data/tier1data/clients/icommands.html). This is the primary way through which data can be downloaded to compute instances on the VSC clusters.

Downloading the downscaled videos

To download the `640x360` videos, install the required Python packages, including `mango_auth`:

```
python -m pip install mango_auth python-irodsclient tqdm
```

Save the script below as `download_kitchen_mango.py` (or download it from the Project folder). Run it from the directory where you want the `EPIC-KITCHENS/videos_640x360` folder created:

```
python download_kitchen_mango.py
```

The script prompts for your R-number to authenticate with ManGO, then downloads the `.mp4` videos into participant subfolders. Files that already exist locally are skipped.

```
import os
import ssl
from irods.session import iRODSSession
from tqdm import tqdm
```

```
from mango_auth import iinit
rnumber = input("Enter your R-number: ")
iinit(rnumber, "set", "set.irods.icts.kuleuven.be")

# Load iRODS environment file
try:
    env_file = os.environ['IRODS_ENVIRONMENT_FILE']
except KeyError:
    env_file = os.path.expanduser('~/.irods/irods_environment.json'

# Configure SSL settings
ssl_context = ssl.create_default_context(purpose=ssl.Purpose.SERVER
ssl_settings = {'ssl_context': ssl_context}

# Define iRODS and local base paths
irods_base_path = "/set/home/ciis-lab/datasets/epic-kitchens/videos
local_base_path = "EPIC-KITCHENS/videos_640x360"

# Start iRODS session
with iRODSSession(irods_env_file=env_file, **ssl_settings) as sessi
    # List participant folders in the iRODS collection
    irods_collection = session.collections.get(irods_base_path)
    participants = [obj.name for obj in irods_collection.subcollect

    for participant in tqdm(participants, desc="Downloading videos"
        participant_local_path = os.path.join(local_base_path, part
        os.makedirs(participant_local_path, exist_ok=True)

        # Get all video files under this participant's collection
        participant_collection_path = f"{irods_base_path}/{particip
        participant_collection = session.collections.get(participan

        for data_obj in tqdm(participant_collection.data_objects, d
            if data_obj.name.endswith('.mp4') or data_obj.name.ends
                local_video_path = os.path.join(participant_local_p
                irods_video_path = f"{participant_collection_path}/

                # Download file if it doesn't exist locally
                if not os.path.exists(local_video_path):
                    session.data_objects.get(irods_video_path, loca
```

EPIC-KITCHENS-100

The dataset is available on ManGO at the following path: `/set/home/ciis-lab/datasets/epic-kitchens`. It contains the `videos` and `annotations` folders as described in the dataset readme.

This downscaled version `640x360` is around `17 GB`.

Evaluation Outputs Format

We expect the evaluation outputs to be in the following format:

```
[
    {
        'verb_output': Iterable of float, shape [97],
        'noun_output': Iterable of float, shape [300],
        'narration_id': str, e.g. 'P01_101_1'
    }, ... # repeated for all segments in the course hidden test se
]
```

Where `verb_output` and `noun_output` are the predicted class probabilities for the 97 verb classes and 300 noun classes respectively, and `narration_id` is the unique identifier for the segment. This format can be saved via `torch.save` with `.pt` suffix.

Calculating Evaluation Metrics

Submit your predictions to Codabench for scoring and detailed feedback. See the competition submission guide for the procedure, metrics, and submission limits.